

Chapter 3: Describing Data: Numerical Measures

Khaled Bastaki

k.bastaki@bhres.group

Measures of Location

Consider a **sample** of Bahraini public schools, in which you have acquired the average math grade of each school:

School i	Average Mathematics Grade x	Governorate
School 1	87	Capital
School 2	58	Muharraq
School 3	46	Muharraq
School 4	48	Southern
School 5	80	Capital
School 6	67	Northern
School 7	75	Capital
School 8	67	Capital
School 9	82	Southern

Measures of Location

We are interested in various ways to calculate the **measures of location**.

This will include various ways to calculate the mean, as well as the median and mode.

Mean

In order to calculate the mean of the **sample**, we apply the formula:

$$\bar{x} = \frac{\sum x_i}{n}$$

where n is the number of observations and $\sum x_i$ is the addition of all grades in your dataset.

Mean

Following from this, we calculate the result as:

$$\bar{x} = \frac{87 + 58 + 46 + 48 + 80 + 67 + 75 + 67 + 82}{9} = 67.7$$

The mean is also called the average of your variable.

Mean

Notice that when calculating the mean, we are using all the data values to come up with the final result.

Although you should keep in mind that the mean is weak when the dataset has extreme values.

Mean

Before leaving this section, you should know that we are dealing with a **sample** of our dataset, the way we denote the average is by using $\bar{x}$.

However, when our dataset is the **population**, then we simply denote it in a different way:

$$\mu = \frac{\sum x_i}{N}$$

The only difference here is that we use the Greek letter μ instead of $\bar{x}$ and that we denote the number of observations with capital N rather than small n .

Mean

Activity:

The *school_math.xlsx* file contains average math scores for all the schools in the country. Use this data to compute the population average.

Note: the excel function to use is `=AVERAGE()`.

Mean

Solution:

On excel, click on any empty cell and type: =AVERAGE().

In the brackets, simply select the entire column of the variable AvgMath.

For $N = 800$, we compute the population mean as:

$$\mu = 71$$

Median

The median is simply the midpoint of the values in the dataset. In order to do this, you would need to arrange the numbers from lowest to highest.

Median

Initially, the dataset shows the following values:

$$87 + 58 + 46 + 48 + 80 + 67 + 75 + 67 + 82$$

Now we re-arrange it from lowest to highest:

$$46 + 48 + 58 + 67 + 67 + 75 + 80 + 82 + 87$$

The median is the value that is in the middle of the dataset. Since there is 9 data points, then the 5th data point is the median, which is 67.

Median

Notice in the current example, there are 9 data points. In the event that we have an odd number of data points, then we simply select the middle data value.

However, if it is an even number of values, then we need to take the average of the middle values.

Measures of Location

Activity: Consider the below sample of car sales for the Toyota showroom:

3000 3500 2100 9400 6400 7450

Find the median of the sample values.

Measures of Location

Solution:

We first arrange the numbers in order:

2100 3000 3500 6400 7450 9400

Since there are 6 data points, which is an even number, then we need to take the 3rd and 4th data point and take its average:

$$\frac{3500 + 6400}{2} = 4950$$

This demonstrates how to get the median for an even numbered sample.

Mode

The mode of a dataset is simply the value that occurs most frequently.

Using our math data from Bahraini schools:

$$87 + 58 + 46 + 48 + 80 + 67 + 75 + 67 + 82$$

In this case, 67 is the mode of this dataset. Note that it is certainly possible to have more than one mode in a dataset or no mode at all.

Weighted Mean

We can mix the approach of the mean and the mode by using the weighted mean.

In this case, we take the average of the sample values but we weight it based on how many observations have the same value.

Weighted Mean

The weighted mean is calculated as:

$$\bar{X}_W = \frac{\sum xw}{w} = \frac{w_1x_1 + w_2x_2 + \dots + w_nx_n}{w}$$

where x_i is the data value while w_i is the amount of times the value appears in the dataset.

Measures of Location

Activity: Akhbar Al Khaleej is a newspaper company and printing press in Bahrain. They have 26 factory workers and they are paid the following hourly wages:

- 14 employees are paid 16.50 BD per hour.
- 10 employees are paid 19.00 BD per hour.
- 2 employees are paid 25.00 BD per hour.

Calculate the weighted mean.

Measures of Location

Solution:

We need to weigh the data values based on how often they appear in the dataset. In this case:

$$\bar{X}_w = \frac{14(16.5) + 10(19) + 2(25)}{14 + 10 + 2} = \frac{471}{26} = 18.12$$

Geometric Mean

Consider the GDP of Bahrain for the past 5 years:

Year	GDP Growth	GDP Growth Ratio
2021	14.0%	0.14
2022	13.7%	0.137
2023	−0.5%	−0.005
2024	2.2%	0.022
2025	3.7%	0.037

Geometric Mean

Let us say we are interested in Bahrain's average GDP growth over these years. It would simply be:

$$\begin{aligned}\bar{x} &= \frac{0.14 + 0.137 - 0.005 + 0.022 + 0.037}{5} \\ &= \frac{0.330}{5} \\ &= 0.066\end{aligned}$$

which means Bahrain's economy grew on average by 6.6%.

Geometric Mean

One thing to consider is that Bahrain's GDP is likely correlated across the years.

This means that if GDP was high in one year, then it might lead it to be high next year as well. If it is low in one year, then it might mean that it will be low in the next year.

In a situation in which a variable is compounded by its past values, then it is better to use the geometric mean to better approximate its average growth.

Geometric Mean

To compute the geometric mean, we need to follow these steps:

- (1) Add 1 to each growth ratio.

Year	GDP Growth	GDP Growth Ratio g	Growth Factor $(1 + g)$
2021	14.0%	0.139	1.139
2022	13.7%	0.137	1.137
2023	-0.5%	-0.005	0.995
2024	2.2%	0.022	1.022
2025	3.7%	0.037	1.037

Geometric Mean

(2) Compute the below calculation

$$= \sqrt[n]{x_1 \cdot x_2 \cdot x_3 \cdot \dots \cdot x_n}$$

where x_i is the data values and n is the number of observations.

Geometric Mean

In this case:

$$\begin{aligned} &= \sqrt[5]{1.139 \cdot 1.137 \cdot 0.995 \cdot 1.022 \cdot 1.037} \\ &= \sqrt[5]{1.366} \\ &= 1.0643 \end{aligned}$$

Geometric Mean

(3) Subtract this value by 1 and convert it into a percentage.

$$= 1.0643 - 1$$

$$= 0.0643$$

$$= 6.43\%$$

Geometric Mean

The interpretation is that Bahrain's average annual growth rate over the years was 6.43%.

We would consider this to be more accurate than the simple mean since it correctly accounts for the compounding over the years.

Measures of Location

Example:

Consider the sales growth of Saudi Aramco's SABIC facility:

Year	Sales Growth
2001	30%
2002	20%
2003	-40%
2004	200%

Calculate the geometric mean using the steps specified in the section.

Measures of Location

Solution:

We start by computing the growth ratio and then use that to add 1 to each observation:

Year	Sales Growth	Growth Ratio g	Growth Factor $(1 + g)$
2001	30%	0.30	1.30
2002	20%	0.20	1.20
2003	-40%	-0.40	0.60
2004	200%	2.00	3.00

Measures of Location

We then compute the result:

$$\begin{aligned} &= \sqrt[4]{1.30 \times 1.20 \times 0.60 \times 3.00} \\ &= \sqrt[4]{2.808} \\ &= 1.2946 \end{aligned}$$

Where we the subtract by 1 and convert it to percentage:

$$\begin{aligned} &= 1.2946 - 1 \\ &= 0.2946 \\ &= 29.46\% \end{aligned}$$

Measures of Dispersion

In statistics, we are often interested in how the data values are spread over the sample. Measures of dispersion therefore allows us to better understand sample's distribution.

Range

Consider the below sample of employees from a factory producing aluminium products in Bahrain and from the UAE:

Bahrain

<i>i</i>	Hourly Production
1	48
2	49
3	49
4	50
5	50
6	50
7	51
8	51
9	52

UAE

<i>i</i>	Hourly Production
1	40
2	47
3	47
4	50
5	50
6	50
7	53
8	53
9	60

Range

We are interested in calculating the range for both these samples. This is simply:

$$\text{Range} = \text{Maximum Value} - \text{Minimum Value}$$

Range

For both factories, we get:

$$\text{BAH} = 52 - 48 = 4$$

$$\text{UAE} = 60 - 40 = 20$$

Range

An interesting thing to keep in mind is that if you took the average production of both factories, you will get a value of 50 for both:

$$\begin{aligned}\bar{x}_{\text{Bahrain}} &= \frac{48 + 49 + 49 + 50 + 50 + 50 + 51 + 51 + 52}{9} \\ &= \frac{450}{9} \\ &= 50\end{aligned}$$

$$\begin{aligned}\bar{x}_{\text{UAE}} &= \frac{40 + 47 + 47 + 50 + 50 + 50 + 53 + 53 + 60}{9} \\ &= \frac{450}{9} \\ &= 50\end{aligned}$$

Range

However, there is a larger **spread** for the UAE factory compared to the Bahraini factory. This is why we are interested in measuring dispersion as it tells us a different story to the data.

Variance and Standard Deviation

Another method to determine the dispersion is by computing the variance of a sample. This is found by using the below formula:

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

Variance and Standard Deviation

We can also use this to compute the standard deviation, which is simply the square root of the variance:

$$S = \sqrt{s}$$

Measures of Dispersion

Activity: Consider again our example with hourly production of aluminium factories:

Bahrain

i	Hourly Production
1	48
2	49
3	49
4	50
5	50
6	50
7	51
8	51
9	52

UAE

i	Hourly Production
1	40
2	47
3	47
4	50
5	50
6	50
7	53
8	53
9	60

Measures of Dispersion

We know that the average of both factories is the same:

$$\bar{x}_{BAH} = \bar{x}_{UAE} = 50$$

Calculate the variance and standard deviation for both factories.

Measures of Dispersion

Solution:

Following from our method:

$$\begin{aligned}s_{BAH}^2 &= \frac{(48 - 50)^2 + (49 - 50)^2 + (49 - 50)^2 + (50 - 50)^2 + (50 - 50)^2 + (50 - 50)^2}{9} \\&\quad + \frac{(51 - 50)^2 + (51 - 50)^2 + (52 - 50)^2}{9} \\&= \frac{4 + 1 + 1 + 0 + 0 + 0 + 1 + 1 + 4}{9} \\&= \frac{12}{9} \\&= 1.33\end{aligned}$$

Measures of Dispersion

$$\begin{aligned}s_{\text{UAE}}^2 &= \frac{(40 - 50)^2 + (47 - 50)^2 + (47 - 50)^2 + (50 - 50)^2 + (50 - 50)^2 + (50 - 50)^2}{9} \\&\quad + \frac{(53 - 50)^2 + (53 - 50)^2 + (60 - 50)^2}{9} \\&= \frac{100 + 9 + 9 + 0 + 0 + 0 + 9 + 9 + 100}{9} \\&= \frac{236}{9} \\&= 26.22\end{aligned}$$

Measures of Dispersion

Similar to what we saw from the range, the Bahraini factory has a much smaller dispersion/variance compared to the UAE factory.

The standard deviation is simply the square root of the variance:

$$S_{BAH} = s_{BAH}^2 = \sqrt{1.33} = 1.15$$

$$S_{UAE} = s_{UAE}^2 = \sqrt{26.22} = 5.12$$

Measures of Dispersion

You should know that the smaller the standard deviation, the closer the data is to its mean, whereas the larger it is, the further away it is from the mean.

Variance and Standard Deviation

Just like the sample mean, you need to know that there is a different way to denote the **population** variance and standard deviation.

Variance and Standard Deviation

The formula for the population variance is slightly different:

$$\sigma^2 = \frac{\sum (x_i - \bar{x})^2}{N}$$

where σ is a Greek letter, but most crucially we divide it by all observations N instead of $n - 1$.

Variance and Standard Deviation

The population standard deviation is the same as before, except we denote it as:

$$\sigma = \sqrt{\sigma^2}$$

Measures of Dispersion

Activity:

Use the *school_math.xlsx* data to calculate the variance and standard deviation of the population of schools.

On excel, the function for population variance is =VAR.P() and the standard deviation is =STDEV.P().

Measures of Dispersion

Solution:

Similar to the previous example, select an empty cell in the excel sheet and input the above command. Include all the data within those brackets by selecting the column.

$$\sigma^2 = 99$$

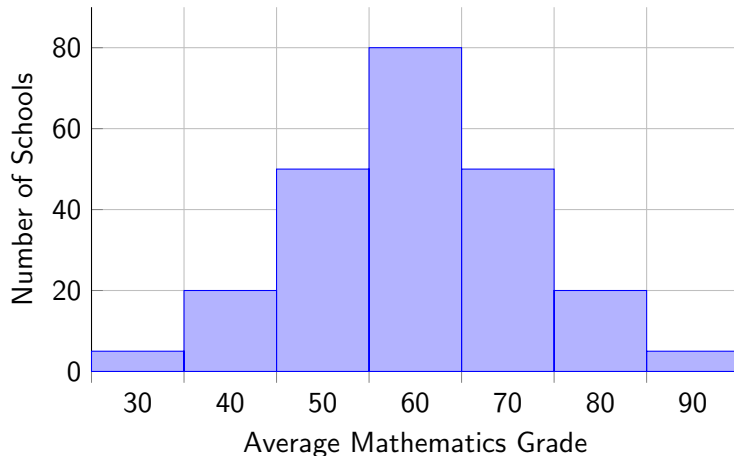
The standard deviation is simply the square root:

$$\sigma = \sqrt{99} = 9.95$$

The Normal Distribution

Consider again our data on average math scores. We can plot the data on a histogram:

Figure: Distribution of Average Mathematics Grades Across Schools



The Normal Distribution

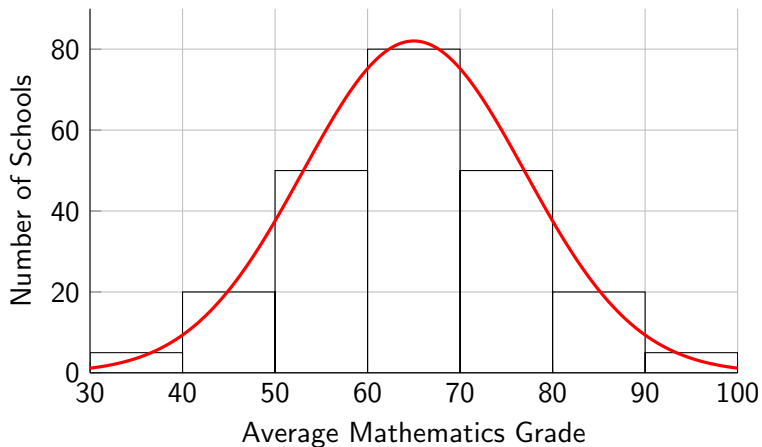
What this shows is that the largest frequency of observations is centred in the middle.

That is, most schools have an average score that is between 60 and 70^a. Schools that have extremely low and extremely high average grades are very few in our observations, which is what we would expect for this dataset.

^aThis is basically what we saw when we computed the mean, where $\bar{x} = 67$.

The Normal Distribution

Figure: Distribution of Average Mathematics Grades Across Schools



The Normal Distribution

This bell-shaped curve is known as the **normal distribution**.

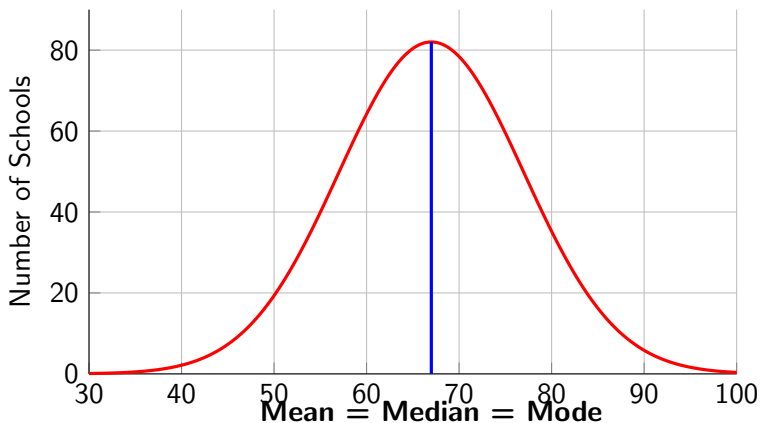
An interesting fact about this fitted line is that the mean, median and mode all happen to be centred around the middle of the distribution.

Recall that when we conducted the measures of location for this dataset, we got:

- Mean = 67.7
- Median = 67
- Mode = 67

The Normal Distribution

which is basically what we are seeing in the graph:



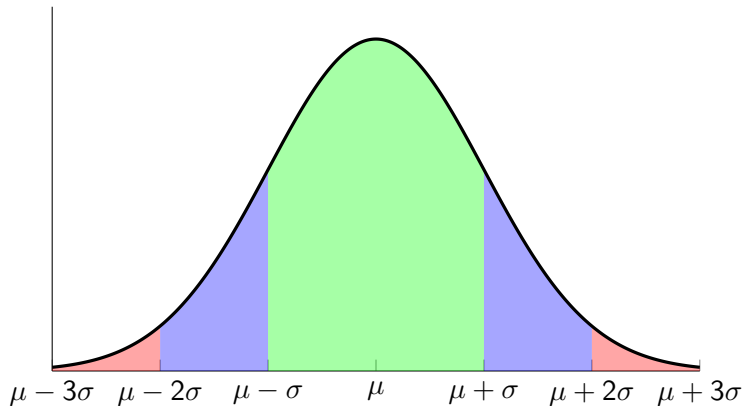
The Normal Distribution

When we are dealing with a normal distribution, it is worth keeping in mind how much of the observations are located around the distribution.

This is more easily understood when considering the standard deviation.

The Normal Distribution

Figure: The Empirical Rule for a Normal Distribution



The Normal Distribution

This illustration shows the following fact:

- The green region represents 68% of all your observations, which is the distance between $\mu - \sigma$ and $\mu + \sigma$.
- Including the blue region represents 95% of all your observations, which is the distance between $\mu - 2\sigma$ and $\mu + 2\sigma$.
- Including the red region represents 99.7% of all your observations, which is the distance between $\mu - 3\sigma$ and $\mu + 3\sigma$.

The Distribution of the Data

Activity

The average mathematics grades across schools are approximately symmetric and bell-shaped. The mean mathematics grade is 67, and the standard deviation is 10.

Using the Empirical Rule, find the range of average mathematics grades that contains approximately:

- 68% of schools.
- 95% of schools.
- 99% of schools.

Illustrate the result in a plot with a normal distribution.

The Distribution of the Data

Solution:

$$68\%: \quad 67 \pm 1(10) = 57 \text{ to } 77$$

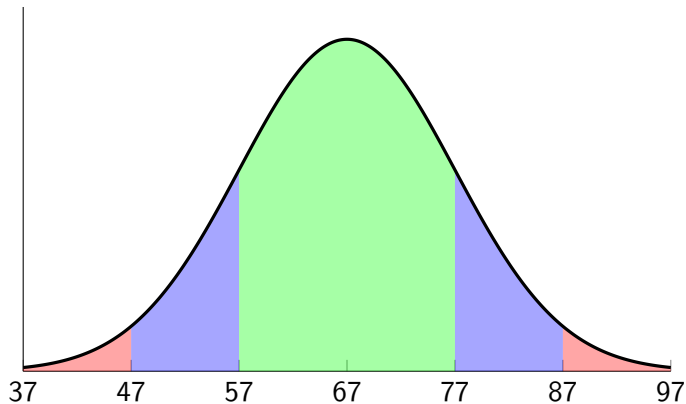
$$95\%: \quad 67 \pm 2(10) = 47 \text{ to } 87$$

$$99.7\%: \quad 67 \pm 3(10) = 37 \text{ to } 97$$

The Distribution of the Data

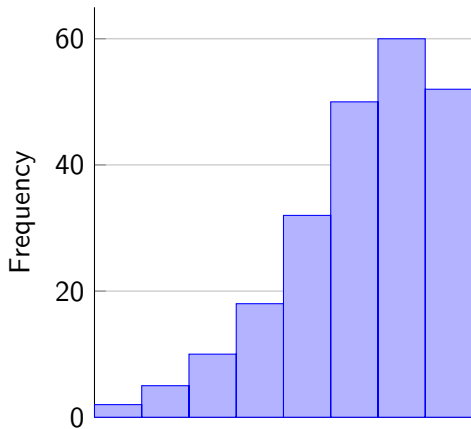
The corresponding illustration:

Figure: The Empirical Rule for a Normal Distribution of Math Scores

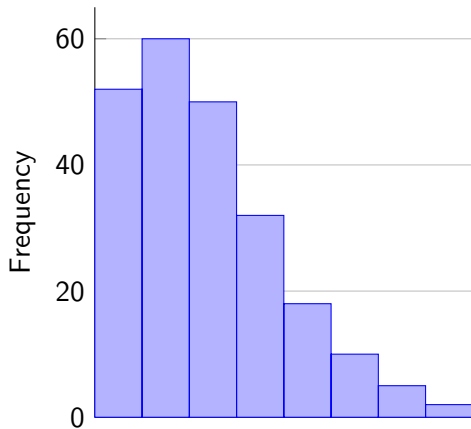


Skewness

Not all data is normally distributed. Sometimes, data can be skewed negatively or positively:



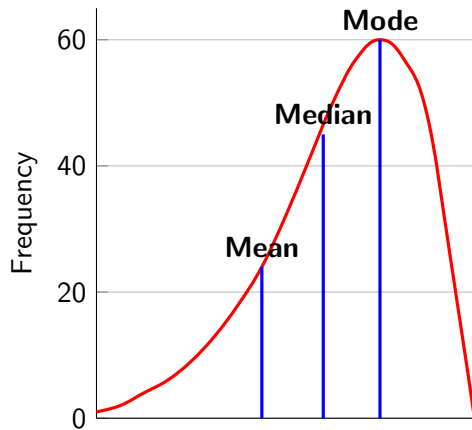
Negatively-Skewed Distribution



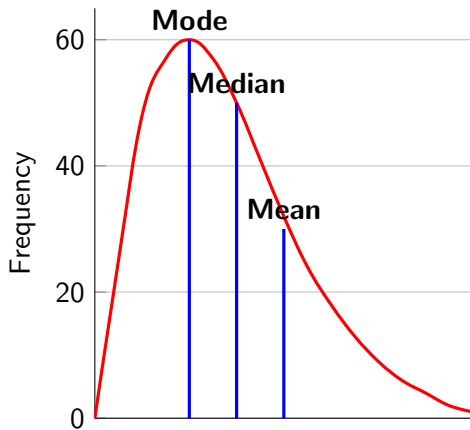
Positively-Skewed Distribution

Skewness

However do note that in these scenarios, the measures of location are not equal:



Negatively-Skewed Distribution



Positively-Skewed Distribution

Skewness

The problem is that when the distribution is not normal, we cannot use the empirical rule to determine where the percentage of observations are located^b.

In this scenario, it is possible to use the **Chebyshev's Theorem** to determine the proportion of values within some k standard deviations around the mean.

^bThat is, when we have a non-normal distribution, we cannot say that 68% of observations are around 1 standard deviation from the mean, and 95% of observations are 2 standard deviations from the mean, etc.

Skewness

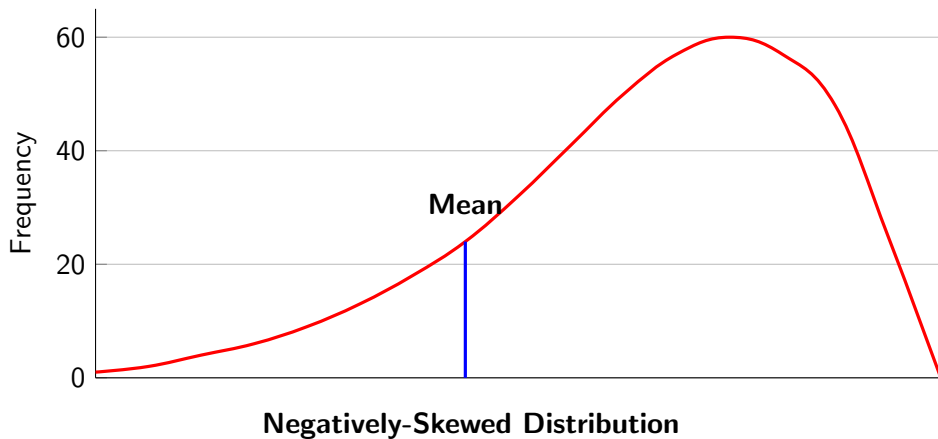
The Chebyshev Theorem is simply:

$$1 - \frac{1}{k^2}$$

where k is the standard deviation. By inserting any value of $k > 1$, you will get the approximate percentage amount of observations that is included in that standard deviation.

Skewness

Consider our negatively-skewed distribution (which is clearly not normally distributed):



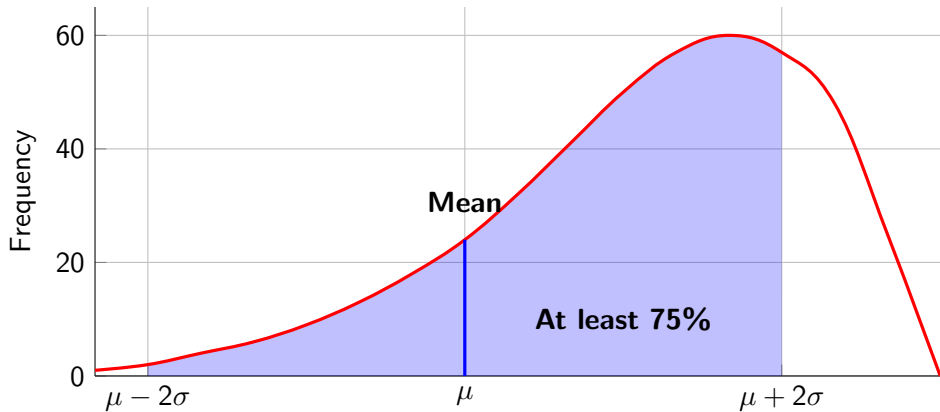
Skewness

Let us saying we are interested to know the amount of observations that are located in 2 standard deviations around the mean. This means $k = 2$:

$$1 - \frac{1}{2^2} = 0.75$$

which means that 75% of the data lies between 2 standard deviations of the distribution.

Skewness



Negatively-Skewed Distribution

The Distribution of the Data

Activity:

Determine the amount of observations located around 3 standard deviations from the mean assuming a non-normal distribution.

The Distribution of the Data

Solution:

When the distribution is not normal, we use Chebyshev's theorem. Since we want to know how many observations are located around 3 standard deviations from the mean, we set $k = 3$:

$$1 - \frac{1}{3^2} = 0.889$$

which means 88.9% of the data is located within 3 standard deviations.